FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Advanced)-2021

PAPER –1

TEST DATE: 13-06-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. K.E. = $\frac{1}{2}m\omega^2x^2$

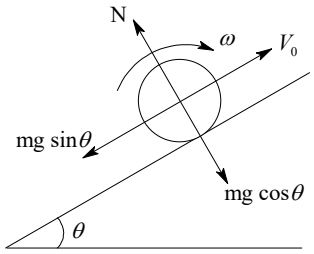
P.E. = $\frac{1}{2}m\omega^2\left(A^2-x^2\right)$

By putting the value of $x = A/\sqrt{2}$, we find that K.E. = P.E.

2. A

Sol. Retarding force for linear velocity = mg $\sin\theta$. All the forces pass through the centre. There is no force to provide moment to reduce angular velocity. So angular velocity will remain constant up to the highest point. The sphere will stop going up at a point where translational velocity becomes zero.

$$\frac{m\left(V_0-0\right)}{t}=mg\sin\theta \text{ or } t=V_0/g\sin\theta$$



3. A

4. C

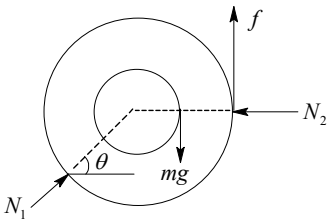
5. B

6. B

7. AD

Sol. $mgr = fR \ --(i)$

$N_1 \sin\theta + f = mg --(ii)$



$N_1 \cos\theta = N_2 ---(iii)$

Solve $(i),(ii)\&(iii)$ to get N_2 and f

So, force at $B = \sqrt{f^2 + N_2^2}$

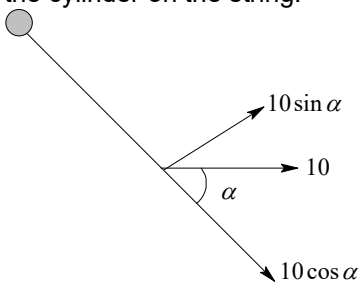
For minimum value of N, use $f \le \mu N_2$

8. ABC

Sol. Use $g_{eff} = g\sin 37 = \dfrac{3g}{5}$ instead of g in all standard vertical circle cases

9. ABD

Sol. The impulse exerted by the string on the cylinder is equal and opposite to the impulse exerted by the cylinder on the string.



$\vec{J} = 40000 \times 0.25 \times 10^{-3} = 10\,ms$ along the string

For marble,

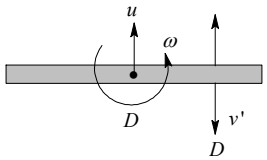
$J = 0 - (-2 \times 10\cos\alpha)$

$\because$ final velocity along the string is zero.

Putting $J = 10ms,$ we get $\alpha = 60°$

10. ABC

Sol. The ball has v' component of its velocity perpendicular to the length of rod immediately after collision u is the velocity of CM of the rod and ω is the angular velocity of the rod just after collision.

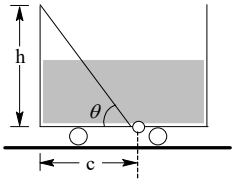


Write velocity of separation = velocity of approach for the point of collision.

Use, conservation of linear momentum perpendicular to the rod and conserve angular momentum about point D

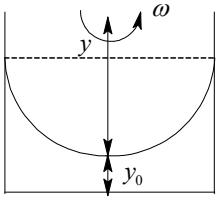
11. AC

Sol. $\tan\theta = \frac{h}{c} = \frac{a}{g}$



12. ABC

Sol. The equation of free surface is $y = \frac{\omega^2 r^2}{2g} + y_0$



SECTION – C

13. 00008.00

14. 00002.20

Sol. $t = \frac{T}{2} = \frac{\pi m}{qB} = \frac{\left(\frac{22}{7}\right)\times\left(14\times10^{-3}\right)}{\left(3\times10^{-6}\right)\left(200/3\right)} = 220s$

15. 00006.00

Sol. $R = \frac{2T}{\rho gh} = \frac{2\times73\times10^{-3}}{10^3\times9.8\times25\times10^{-3}} \qquad = 0.6\times10^{-3}m > 0.25mm$

16. 00001.00

Sol. $M = \sqrt{2}\times i.\frac{\pi r^2}{2} \qquad = \sqrt{2}\times i.\frac{\pi}{2}\times\left(\frac{l}{2\pi}\right)^2 \quad = \frac{\sqrt{2}iL^2}{8\pi}$

17. 00005.00

Sol. $y = 8\sin\left\{\pi\left(f_1 - f_2\right).t\right\}\cos\left\{\pi\left(f_1 + f_2\right)t\right\}$

18. 00000.25

Chemistry

PART – II

SECTION – A

19. D

20. D

21. D

22. B

23. A

24. A

25. ABD

26. ABCD

27. ABD

28. BC

29. BC

30. BC

SECTION – C

31. 00000.22

Sol. $3.7\times10^{10}=\dfrac{0.693}{5730\times365\times24\times60\times60}\times\text{No. of atoms}$

$\therefore$ number of atoms $=9.65\times10^{21}$

$\because$ 6.023×10^{23} atoms of ^{14}C = 14g

$\therefore$ 9.65×10^{21} atoms of $^{14}C=\dfrac{14\times9.65\times10^{21}}{6.023\times10^{23}}=0.2243\,g$

32. 00006.67

Sol. Total moles $=2+2+2=6$

$p_{PCl_5}=2\times3/6=1,\ p_{PCl_3}=1,\ p_{PCl_2}=1$

$K_p=\dfrac{(1)(1)}{1}=1\,atm$

When volume is double (i.e. $V\rightarrow2V$) at constant P and T, total moles become = 12.
If volume is doubled, reaction will more backward at constant P and T.

$p_{PCl_5}=\dfrac{(2+y)}{12}\times3,\ p_{PCl_3}=\dfrac{(2-y)}{12}\times3$

$p_{Cl_2}=\dfrac{(2+x)-y}{12}\times3$

$$\therefore K_p = 1 = \frac{\left(\frac{2-y}{12}\right)\times 3\times\left[\frac{(2+x)-y}{12}\right]\times 3}{\left(\frac{2+y}{12}\right)\times 3} \quad .....(1)$$

Also $2+y+2-y+(2+x)-y=12$

From equations (i) and (ii), solve y = 2/3, x = 20/3

33. 00002.00

Sol. Meq. Of acid in 25 mL $=38.5\times 1$

$\therefore$ Meq. Of acid in $1000\,\text{mL}=\frac{38.5\times 1\times 1000}{25}=1540$

$\therefore$ Normality of acid $=\frac{1540}{1000}=1.54$

Also, molality $=\frac{\Delta T}{K_f}=\frac{0.4}{0.52}=\text{molarity}$

$$\frac{0.4}{0.52}=\frac{1.54}{n}$$

$$n=2$$

34. 00001.84

Sol. Given:

$Fe^{6+}+3e\rightarrow Fe^{3+}; \qquad -\Delta G_1^\circ = 3\times 2.20\times F \quad .....(i)$

$Fe^{3+}+e\rightarrow Fe^{2+}; \qquad -\Delta G_2^\circ = 1\times 0.77\times F \quad .....(ii)$

$Fe^{2+}+2e\rightarrow Fe^{\circ}; \qquad -\Delta G_3^\circ = 2\times(-0.445)\times F .....(iii)$

By eq. (i) and (ii)

$Fe^{6+}+4e\rightarrow Fe^{2+}; \qquad -\Delta G_4^\circ = -\Delta G_1^\circ + (-\Delta G_2^\circ)$

$\because 4\times E_4^\circ\times F = 3\times 220\times F + (1\times 0.77\times F)$

$\therefore E_4^\circ = 1.84V$

35. 00004.33

Sol. Equivalent of Ag deposited $=\frac{1.307}{108}=0.0121$

$\therefore$ Equivalent of Cu^{2+} lost $=0.0121$

Now initial Equivalent of $CuSO_4.5H_2O=\left[10/(249.54/2)\right]=0.0802=\text{Initial eq. of } Cu^{2+}$

$\therefore$ Wt. of Cu^{2+} left$=\frac{0.0681\times 63.54}{2}=2.163g$ in 500 mL

$\therefore$ Wt. of Cu^{2+} left in 1 litre $H_2O=2.163\times 2$

$=4.327\,g\,litre^{-1}$

36. 00007.00

Sol. $Cu^{2+} + e \rightarrow Cu^{+}$

$2I^{-} \rightarrow I_2 + 2e$

$2S_2O_3^{2-} \rightarrow S_4O_6^{-} + 2e$

$\text{Meq. of } Cu^{2+} = \text{Meq. of liberated } I_2$

$= Meq.of\ Na_2S_2O_3 = 12.12 \times 0.1 \times 1 = 1.212$

$\therefore w_{Cu^{2+}} = 0.077g = w_{Cu} \qquad \left(Cu \xrightarrow{H_2SO_4} CuSO_4\right)$

$\therefore \%Cu = \dfrac{0.077}{1.10} \times 100 = 7\%$

Mathematics

PART – III

SECTION – A

37. C

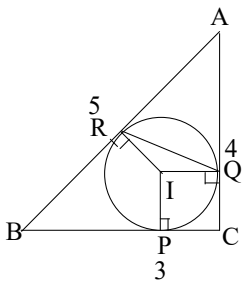
Sol. In Δ IRQ, $\angle IRQ = \pi - A$

$$\frac{RQ}{\sin A} = \frac{r}{\sin\left(\frac{A}{2}\right)}$$

$$RQ = 2r\cos\left(\frac{A}{2}\right),\ RP = 2r\cos\left(\frac{B}{2}\right),\ PQ = 2r\cos\left(\frac{C}{2}\right)$$

$$PR^2 + RQ^2 + QP^2 = 4r^2\left[\cos^2\left(\frac{A}{2}\right) + \cos^2\left(\frac{B}{2}\right) + \cos^2\left(\frac{C}{2}\right)\right]$$

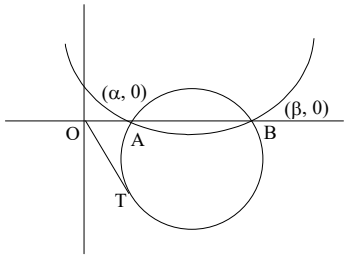
$$= 4r^2\left[\frac{1+\cos A}{2} + \frac{1+\cos B}{2} + \frac{1}{2}\right] = 2r^2\left[3 + \frac{3}{5} + \frac{4}{5}\right] = \frac{44r^2}{5}.$$



38. D

Sol. Use power of a point;

$OT^2 = OA.\ OB = \alpha\beta = \frac{c}{a} \Rightarrow OT = \sqrt{\frac{c}{a}}$.



39. D

Sol. Let $T_{r+1} = {}^{20}C_r\ x^{20-r}(2y)^r = k\,x^a y^b$ and $T_{s+1} = {}^{20}C_s x^{20-s}(2y)^s = kx^{a+1}y^{b-1}$ for given condition r-s = 1 and ${}^{20}C_{1+s}.2^{1+s} = {}^{20}C_s 2^s \Rightarrow s = 13$ then $r = 14 \Rightarrow a = 6$.

40. B

41. D

42. B

Sol. $(x-b)^2 + \sin^2 x$ is minimum

x = 0, b = 2 as $b \in [2,3]$

43. A, D

Sol. Equation of tangent to $y = -x^2$ is $x = my - \frac{1}{4m}$, m > 0

$\Rightarrow x - my + \frac{1}{4m} = 0$

since this line is tangent to $x^2 + (y-2)^2 = 4$

$\Rightarrow m = \frac{1}{4\sqrt{5}}$

Hence, equation of tangent is $y = 4\sqrt{5}x + 20$.

Case II: equation of tangent to $y = x^2$ is $x = my + \frac{1}{4m}$, $m > 0$ this line is tangent to

$x^2 + (y - 2)^2 = 4$

$\Rightarrow m = \frac{1}{4\sqrt{3}}$

equation of tangent is $y = 4\sqrt{3}x - 12$.

44. B, C

Sol. For K = 2, each element has four possibility and three out of four are favorable for the event.

Hence required probability = $\left(\frac{3}{4}\right)^n$.

For K = 3, Probability that intersection is empty is $\left(\frac{7}{8}\right)^n$

For K = 3, Probability that intersection is singleton = $\frac{n}{8}\left(\frac{7}{8}\right)^{n-1}$

45. A, D

Sol. $y = 4\sin^2 x + 3\cos^2 x - 24\sin\frac{x}{2} - 24\cos\frac{x}{2}$

$= 3 + \sin^2 x - 24\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)$

Let $\left(\sin\frac{x}{2} + \cos\frac{x}{2}\right) = t$

$\Rightarrow y = t^4 - 2t^2 - 24t + 4 = f(t)$

$\Rightarrow f'(t) = 4(t - 2)(t^2 + 2t + 3) < 0$

since, $0 \le x \le \frac{\pi}{2} \Rightarrow 1 \le t \le \sqrt{2}$

$\Rightarrow$ f(t) is decreasing function for $1 \le t \le \sqrt{2}$ $y_{max} = 4(1 - 6\sqrt{2})$ and $y_{min} = -21$.

46. A, D

Sol. Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$\Rightarrow x, y, z > 0$ and $x + y + z \le 12$.

Number of solution = ${}^{12}C_9 = {}^{12}C_3 = {}^{11}C_8 + {}^{11}C_9$.

47. C, D

Sol. $f(x) = \begin{cases} \sin x, & x > 0 \\ 0, & x = 0 \\ \frac{\tan x^2}{x^2}, & -1 < x < 0 \end{cases}$

Hence, $\lim\limits_{x\to 0^+} f(x) = 0$ and $\lim\limits_{x\to 0^-} f(x) = 1$.

48. A, B, D

Sol. $f(x) = \begin{cases} 5x & x \ge -\frac{1}{2} \\ x - 2 & x < -\frac{1}{2} \end{cases}$ $g(x) = \begin{cases} \frac{x}{5} & x \ge -\frac{5}{2} \\ x + 2 & x < -\frac{5}{2} \end{cases}$

After calculation

$f(g(x)) = x = gf(x)$

$\Rightarrow f = g^{-1}$ and $g = f^{-1}$

Hence $(fof)^{-1} = \left(g^{-1}og^{-1}\right)^{-1} = gog$

And $(fog)^{-1} = gof$.

SECTION – C

49. 00017.00

50. 00002.00

51. 01600.00

Sol. D = 9 + 4a = odd number = perfect square

$\Rightarrow (2k + 1)^2 = 9 + 4a_i$, $k \in I$

$\Rightarrow (4k^2 + 4k + 1) = 9 + 4a_i$

$\Rightarrow 4(k^2 + k) = 4(a_i + 2) \Rightarrow k(k + 1) - 2 = a_i$

Thus a_i = 4, 10, 18, 28, 40, ……

So general term $t_r = r^2 + 3r$

$T_{15} = 270$, $t_{16} = 304$

$\Rightarrow \sum_{i=1}^{15} a_i = \sum_{i=1}^{15} r^2 + 3\sum_{i=1}^{15} r = 1600$

52. 00137.00

Sol. $x = x_1 + x_3 + x_5 + x_7 + x_9$

$Y = x_2 + x_4 + x_6 + x_8$

$(x - y) = 11k$, $k \in z^+$ of $x - y = 11$

and $x + y = 45$

ie x = 28, y = 17 total numbers $(3 \times 4! \times 5!) + (6 + 4! + 5!)$

of $y - x = 1$

$y + x = 45$

ie x = 17, y = 28 total numbers $2 \times 4! \times 5!$

Then all such 9 digit numbers are 1!.4!.5! Required probability $\frac{p}{q} = \frac{1!.4!.5!}{9!} = \frac{11}{126}$

53. 00006.00

Sol. $\cos\pi/7$ is the root of the equation

$\Rightarrow n = 6$

54. 00003.00